Visualization with Genome Browsers

Session 5

T6: Making the most of cis-regulatory annotations for genome-scale analysis

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Session map

IGV
- Definition
- Availability

Using IGV
- Loading/selecting reference genome
- Loading tracks
- Tracks customization

Hands-on

Have you used genome browsers before?

If you have used genome browser before, which one(s) do you prefer the most?

Using genome browsers to visualise annotation

CGGCCTTTGGGCTCGGCCTTCAGCTCAAGACTTAACTTCCCTCCCAGCTGTCCCAGATGACGCCATCTGAAATTTCTTGGAAACACGATCAG

Regulatory element

TTTAACGGAATATTGCTGTTTTGGGGAAGTGTTTTACAGCTGCTGGGCACGCTGTATTTGCCTTACTTAAGCCCCTGGTAATTGCTGTATTG

CGA**AGACATGCTGATGGGAATTA**CCAGGCGGCGTTGGTCTCTAACTGGAGCCCTCTGTCCCCACTAGCCACGCGTCACTGGTTAGCGTGATT

GAAACTAAATCGTATGAAAATCCTCTTCTCTAGTCGCACTAGCCACGTTTCGAGTGCTTAATGTGGCTAGTGGCACCGGTTTGGACAGCACA

GCTGTAAAATGTTCCCATCCTCACAGTAAGCTGTTACCGTTCCAGGAGATGGGGACTGAATTAGAATTCAAACAAATTTCCAGCGCTTCTGA

Gene

GTTTTACCTCAGTCACATAATAAGGAATGCATCCCTGTGTAAGTGCATTTTTGGTCTTCTGTTTTGCAGACTTATTTACCAAGCATTGGAGGA

ATATCGTAGGTAAAA**ATGCCTATTGGATCCAAAGAGAGGCCAACATTTTTTGAAATTTTTAAGACACGCTGCAACAAAGCAGGTATTGACA**

AATTTTATATAACTTTATAAATTACACCGAGAAAGTGTTTTCTAAAAAATGCTTGCTAAAAACCCAGTACGTCACAGTGTTGCTTAGAAC

CATAAACTGTTCCTTATGTGTGTATAAATCCAGTTAACAACATAATCATCGTTTGCAGGTTAACCACATGATAAATATAGAACGTCTAGT

GGATAAAGAGGAAACTGGCCCCTTGACTAGCAGTAGGAACAATTACTAACAAATCAGAAGCATTAATGTTACTTTATGGCAGAAGTTGTC

CAACTTTTTGGTTTCAGTACTCCTTATACTCTTAAAAATGATCTAGGACCCCCGGAGTGCTTTTGTTTATGTAGCTTACCATATTAGAAA

TTTAAAACTAAGAATTTAAGGCTGGGCGTGGTGGCTCACGCCTGTAATCCCAGCACTTTGGGAGGCCGAGGTGGGCGGATCACTTGAGGC

CAGAAGTTTGAGACCAGCCTGGCCAACATGGTGAAACCCTATCTCTACTAAAAATACAAAAAATGTGCTGCGTGTGGTGGTGCGTGCCTGT

AATCCCAGCTACACGGGAGGTGGAGGCAGGAGAATCGCTTGAACCCTGGAGGCAGAGGTTGCAGTGAGCCAAGATCATGCCACTGCACTCTA

GCCTGGGCCACATAGCATGACTCTGTCTCAAAACAAACAAACAAACAAAAAACTAAGAATTTAAAGTTAATTTACTTAAAAATAATGAAAGG

TAACCCATTGCATATTATCACAACATTCTTAGGAAAAATAACTTTTTGAAAACAAGTGAGTGGAATAGTTTTACATTTTTGCAGTTCTCTT

TAATGTCTGGCTAAATAGAGATAGCTGGATTCAACTTATCTGTGTCTAATCTGTTATTTTGGTAGAAGTATGTGAAAAAAATTAACCTCACC

Variant

TTGAAAAAAGGAATATTTTAATAGTTTTCAGTTTACTTTTTGGTATTTTTCCTTGTACTTTGCATAGATTTTTCAAAGATCTAATAGATATAG

CATAGGTCTTTCC**C**ATGTCGCAACATCATGCAGTGATTATTTGGAAGATAGTGGTGTTCTGAATTA

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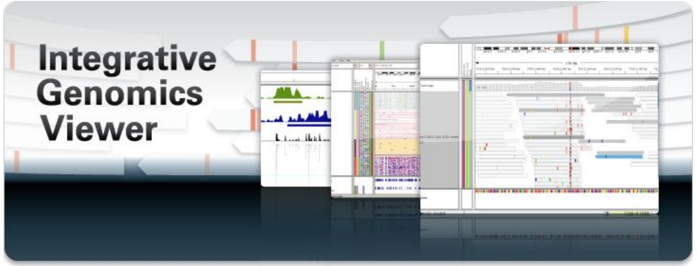
Genome browsers

- ✓ They integrate and allow you to search, browse, retrieve, compare and analyse genomic data.
- ✓ They put genes, regulatory elements and experimental evidence on the same genomic map.
- ✓ They help with the interpretation of regulatory variants and support better experimental hypotheses.

The Integrative Genomics Viewer (IGV)

✓ Is a high-performance, easy-to-use, interactive tool for the visual exploration of genomic data.

✓ It supports flexible integration of all the common types of genomic data and metadata, investigator-generated or publicly available, loaded from local or cloud sources.



✓ Is available in multiple forms, including:

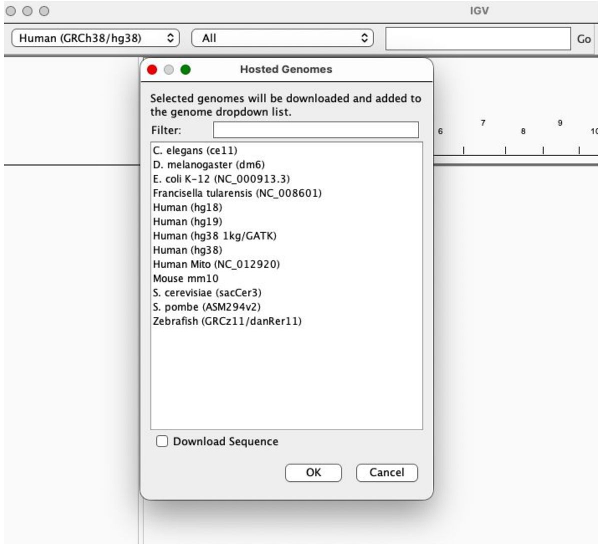
- The original IGV - a Java desktop application,
- IGV-Web - a web application,
- igv.js - a JavaScript component that can be embedded in web pages (for developers)

✓ Is focused on the IGV desktop application.

https://igv.org/doc/desktop/#

Selecting genomes on IGV

✓ The genome dropdown menu on the left end of the IGV window toolbar displays the current reference genome, and contains a selection list of recently loaded genomes.



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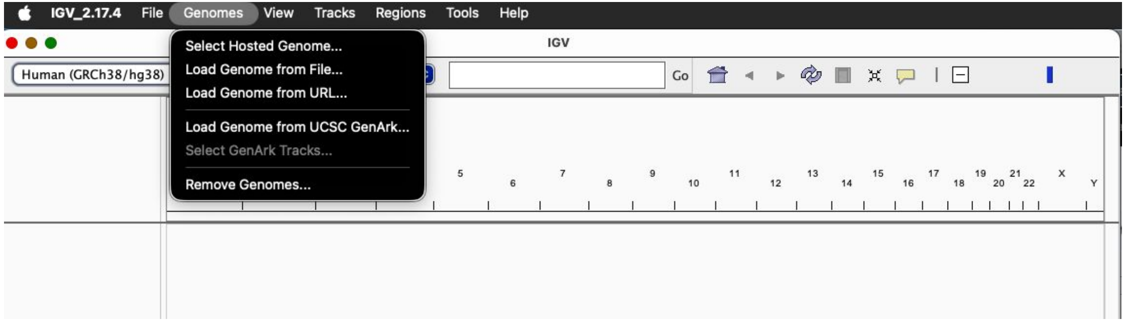
Loading new genomes on IGV

✓ Options for changing loading a genome not yet on the selection list:

- Selecting a predefined genome from the "hosted genome" list
- Loading a UCSC track hub
- Loading a custom genome json or sequence file:
    - JSON: allow to load sequence and annotation
    - FASTA: genome sequence, must be indexed

✓ When you switch to a different reference genome, IGV will clear the current session.



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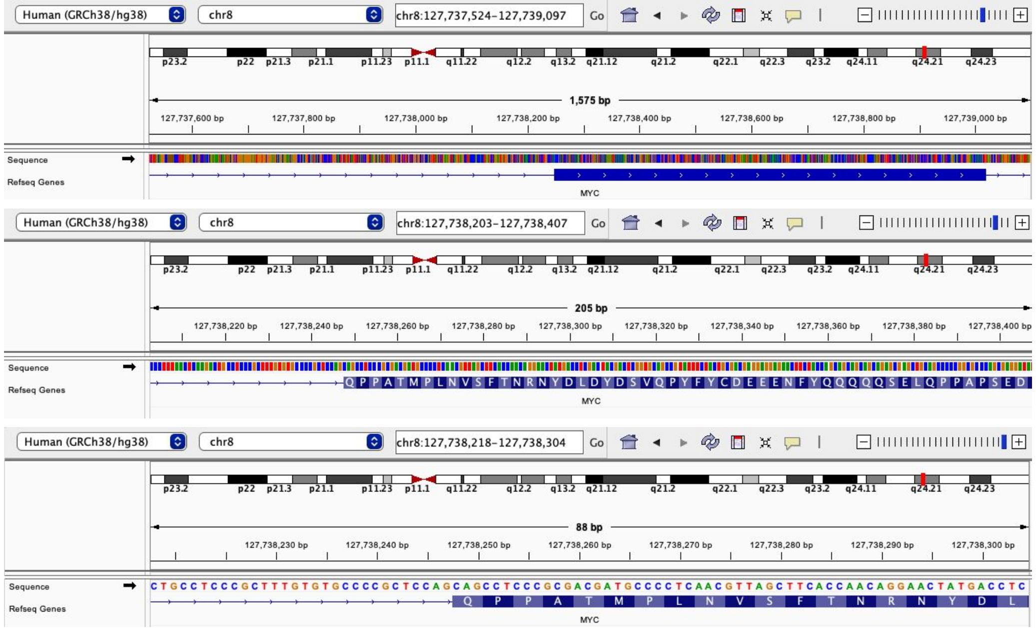
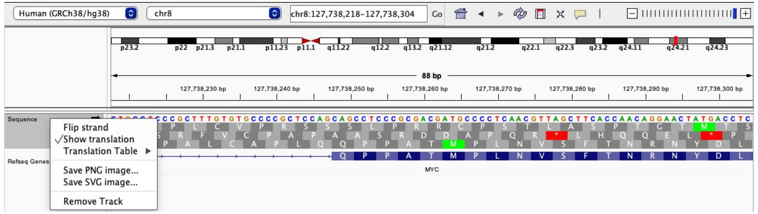
Exploring the genome sequence on IGV

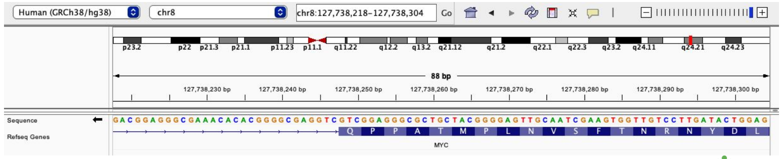
Zoom-in



Show translation



Change strand



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Loading tracks

✓ Tracks can be loaded from local files, URLs, and remote track servers

✓ File formats include:

- BAM (SAM): Must be sorted and indexed
- BED - bigBed
- bedGraph
- bigWig
- narrowPeak, broadPeak (ENCODE)
- GFF/GTF
- VCF

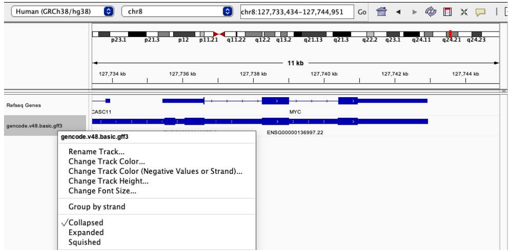
✓ Common attributes:

- Name
- Color
- Height

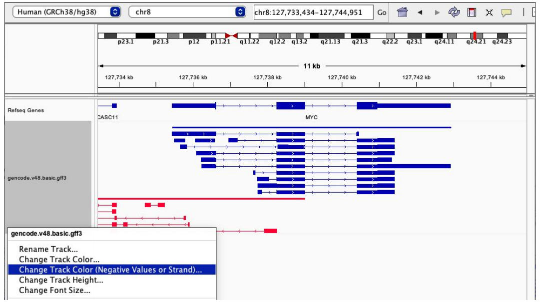
Loading and displaying gene annotation

✓ Formats: bed, bigBed, gtf, gff, psl, and narrowPeak and broadPeak.

✓ Display options:



✓ Group and color by strand:



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Loading and displaying gene annotation

✓ Exploring feature details

Loading peak tracks

✓ Track display options

✓ Change name and color

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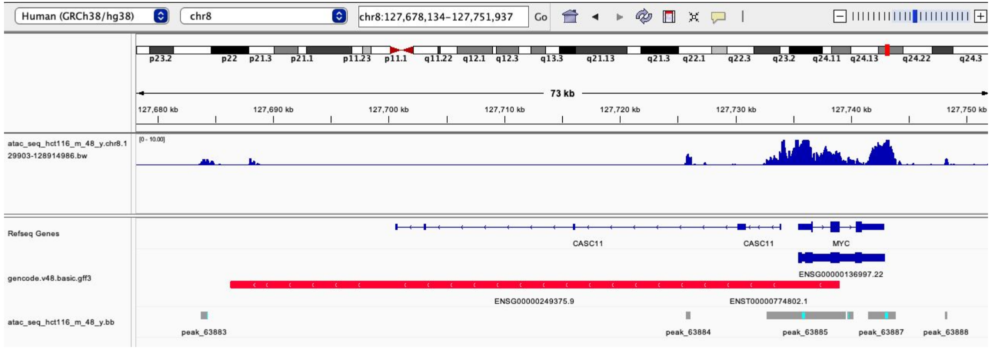
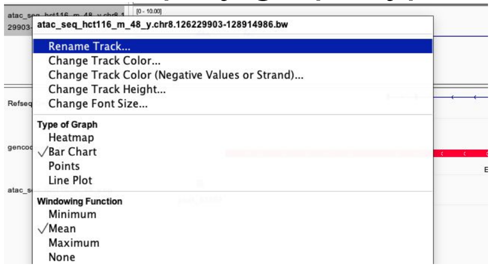
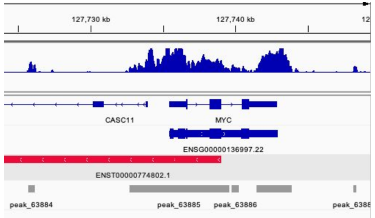
Loading signal tracks

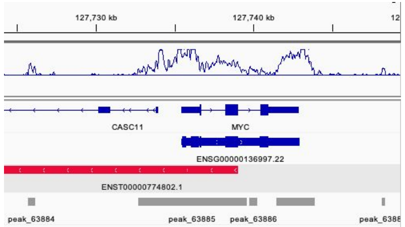
✓ Quantitative tracks file format include: wig, bigWig, bedGraph

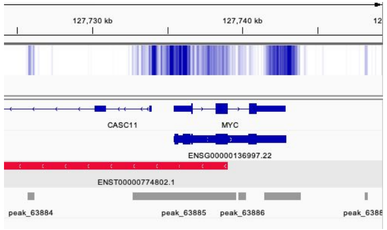


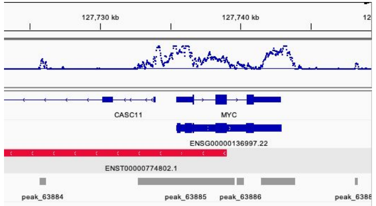
✓ Display graph type











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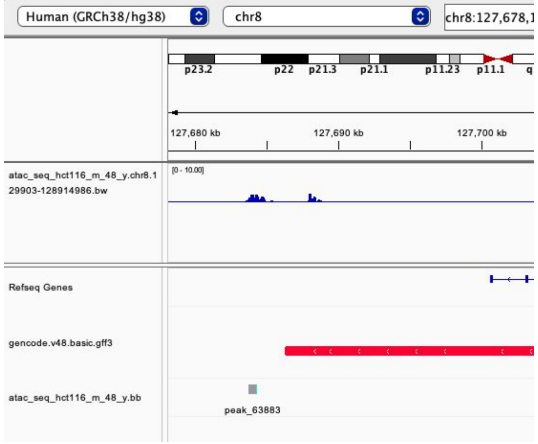
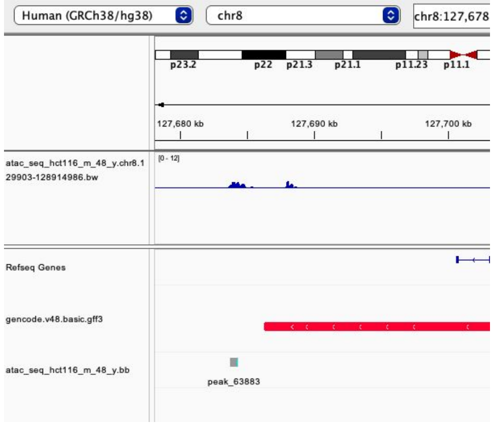
Loading signal tracks

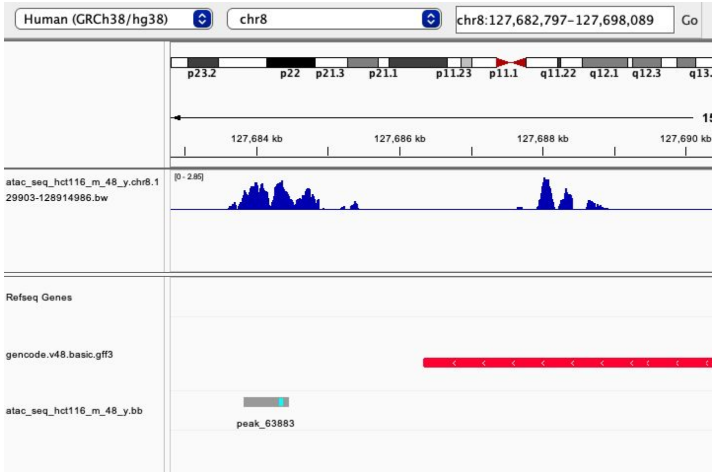
✓ Data range: can be fixed or variable



Fixed

Autoscale





Autoscale + zoom-in

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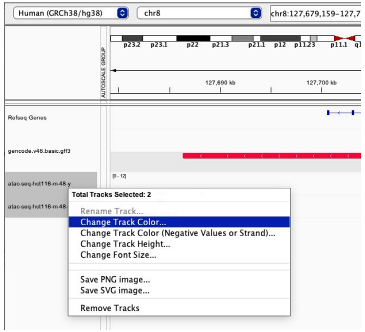
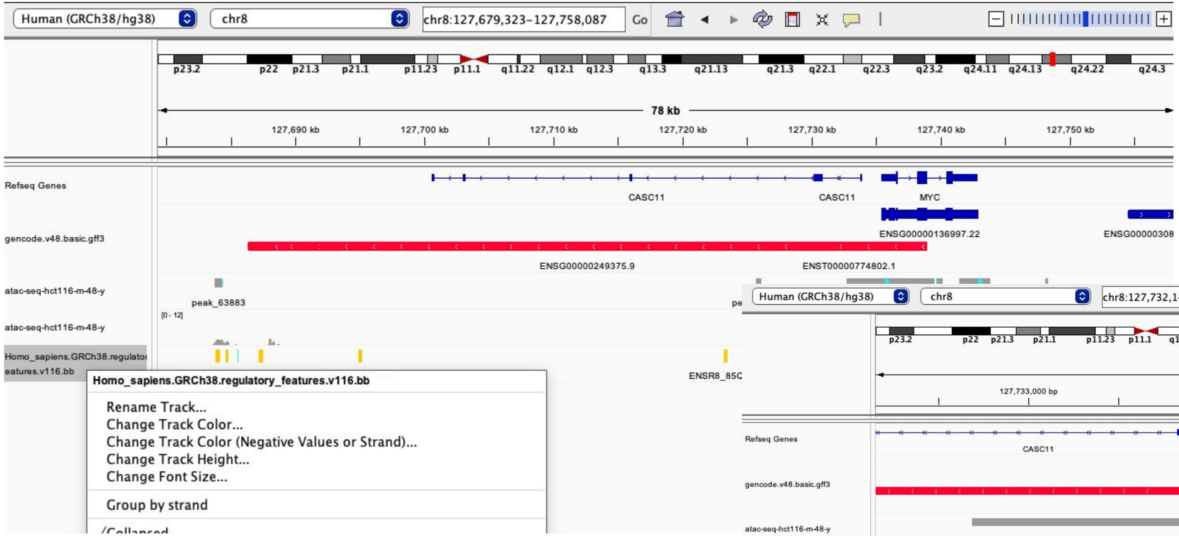
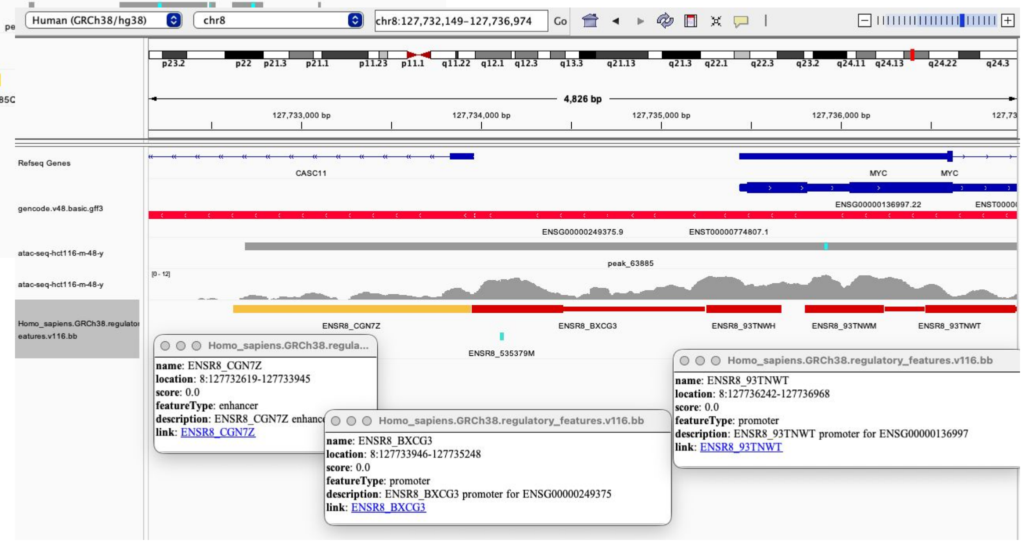
Loading signal tracks

✓ Change name and color



Loading regulatory features track





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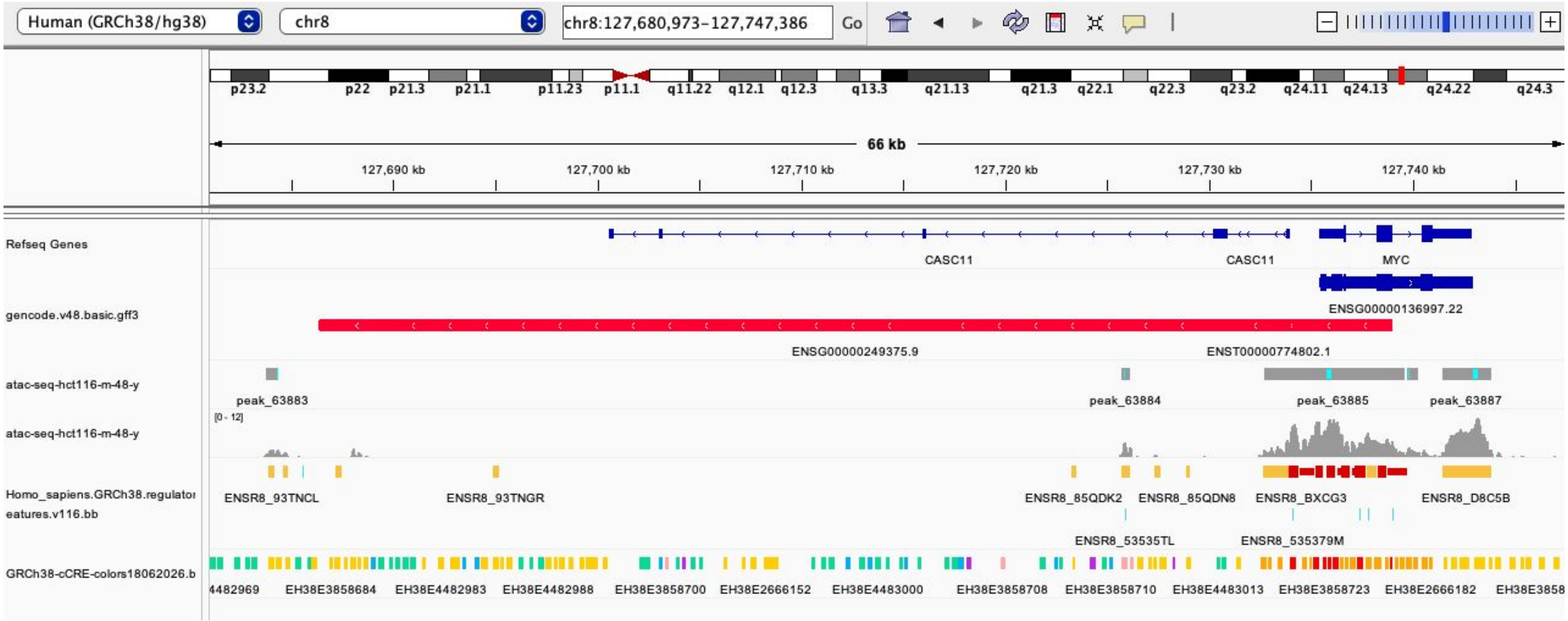
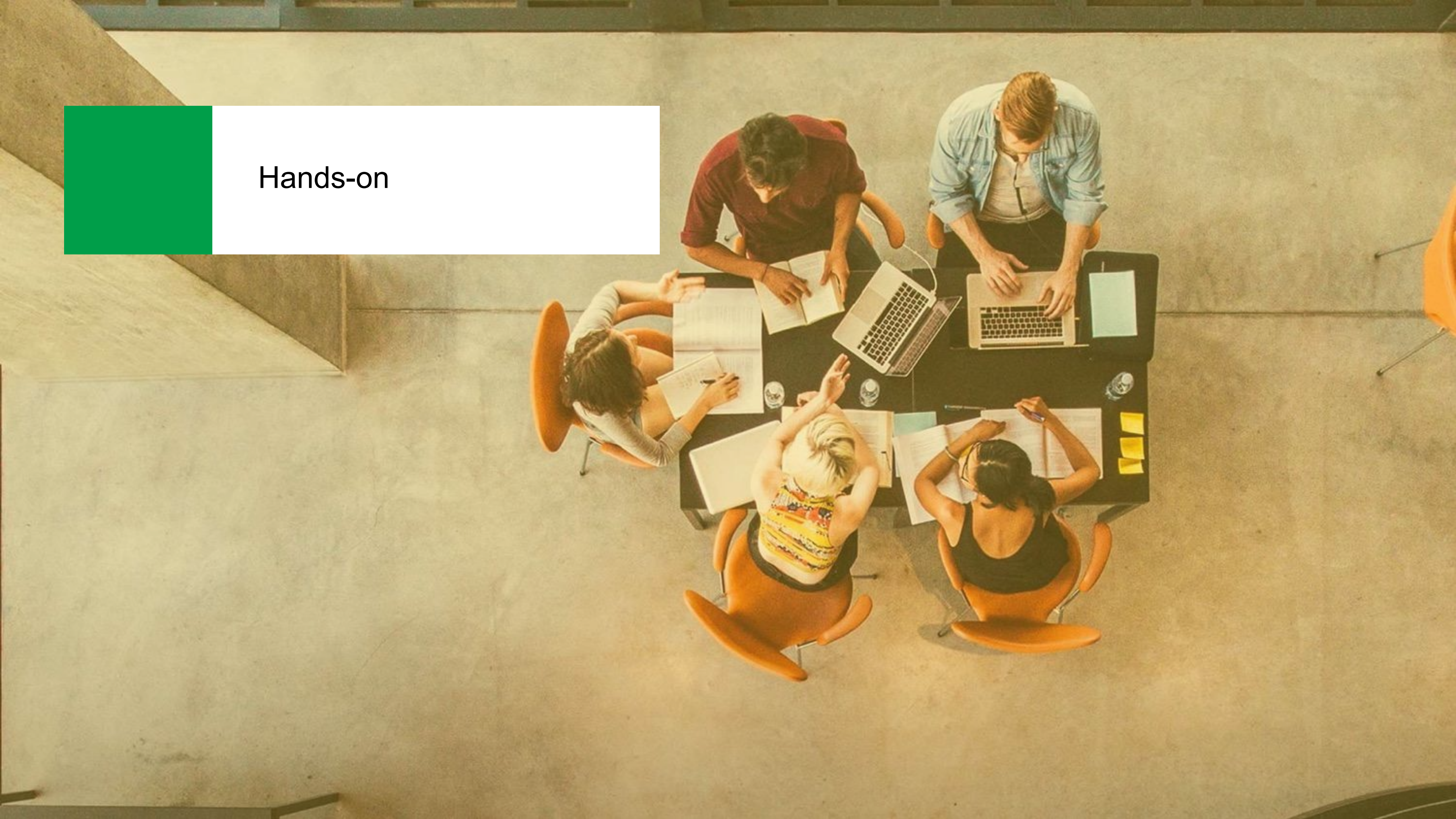
Loading regulatory features track



Questions?

Hands-on



Before starting… files provenance

Table1

	Folder	File name	Source file name	Download date	Origin repository	Source	Version	Assembly
2	ENCODE	ENCFF345CWR.atac.bed	ENCFF345CWR.bed	23/07/2026	ENCODE	https://www.encodeproject.org/experiments/ENC	ENCODE4 v1.8.0	mm10
3	ENCODE	ENCFF633ETU.atac.chr6.5911252-7710120.bw	ENCFF633ETU.bigWig	23/07/2026	ENCODE	https://www.encodeproject.org/experiments/ENC	ENCODE4 v1.8.0	mm10
4	ENCODE	ENCFF053ITU.k27ac.bed	ENCFF053ITU.bed	23/07/2026	ENCODE	https://www.encodeproject.org/experiments/ENC	ENCODE4 v1.5.1	mm10
5	ENCODE	ENCFF607CHT.k27ac.chr6.5911252-7710120.bw	ENCFF607CHT.bigWig	23/07/2026	ENCODE	https://www.encodeproject.org/experiments/ENC	ENCODE4 v1.5.1	mm10
6	ENCODE	ENCFF079TUC.k4me3.bed	ENCFF079TUC.bed	23/07/2026	ENCODE	https://www.encodeproject.org/experiments/ENC	ENCODE4 v1.5.1	mm10
7	ENCODE	ENCFF798ETE.k4me3.chr6.5911252-7710120.bw	ENCFF798ETE.bigWig	23/07/2026	ENCODE	https://www.encodeproject.org/experiments/ENC	ENCODE4 v1.5.1	mm10
8	ENCODE	ENCFF217ZPW.k4me1.bed	ENCFF217ZPW.bed	23/07/2026	ENCODE	https://www.encodeproject.org/experiments/ENC	ENCODE4 v1.6.1	mm10
9	ENCODE	ENCFF013ALT.k4me1.chr6.5911252-7710120.bw	ENCFF013ALT.bigWig	23/07/2026	ENCODE	https://www.encodeproject.org/experiments/ENC	ENCODE4 v1.6.1	mm10
10	ENCODE	ENCFF240LRP.dnase.bed	ENCFF240LRP.bed	20/05/2026	ENCODE	https://www.encodeproject.org/experiments/ENC	ENCODE4 v3.0.0-alpha.2	GRCh38
11	ENCODE	ENCFF906MKW.dnase.chr8.126229903-128914986.bw	ENCFF906MKW.bigWig	20/05/2026	ENCODE	https://www.encodeproject.org/experiments/ENC	ENCODE4 v3.0.0-alpha.2	GRCh38
12	ENCODE	ENCFF296ZZB.atac.bed	ENCFF296ZZB.bed	20/05/2026	ENCODE	https://www.encodeproject.org/experiments/ENC	ENCODE4 v1.9.1	GRCh38
13	ENCODE	ENCFF624HRW.atac.chr8.126229903-128914986.bw	ENCFF624HRW.bw	20/05/2026	ENCODE	https://www.encodeproject.org/experiments/ENC	ENCODE4 v1.9.1	GRCh38
14	ENCODE	ENCFF711MPL.k4me3.bed	ENCFF711MPL.bed	20/05/2026	ENCODE	https://www.encodeproject.org/experiments/ENC	ENCODE4 v1.5.1	GRCh38
15	ENCODE	ENCFF394IMO.k4me3.chr8.126229903-128914986.bw	ENCFF394IMO.bw	20/05/2026	ENCODE	https://www.encodeproject.org/experiments/ENC	ENCODE4 v1.5.1	GRCh38
16	ENCODE	ENCFF349LKU.k27ac.bed	ENCFF349LKU.bed	20/05/2026	ENCODE	https://www.encodeproject.org/experiments/ENC	ENCODE4 v1.5.1	GRCh38
17	ENCODE	ENCFF277XII.k27ac.chr8.126229903-128914986.bw	ENCFF277XII.bigWig	20/05/2026	ENCODE	https://www.encodeproject.org/experiments/ENC	ENCODE4 v1.5.1	GRCh38
18	ENCODE	ENCFF931YSQ.k4me1.bed	ENCFF931YSQ.bed	20/05/2026	ENCODE	https://www.encodeproject.org/experiments/ENC	ENCODE4 v1.6.1	GRCh38
19	ENCODE	ENCFF182EJH.k4me1.chr8.126229903-128914986.bw	ENCFF182EJH.bw	20/05/2026	ENCODE	https://www.encodeproject.org/experiments/ENC	ENCODE4 v1.6.1	GRCh38
20	Ensembl_Regulation	atac_seq_forebrain_m_13.5_d.bb	atac_seq_forebrain_m_13.5_d.bb	24/07/2026	Ensembl Regulation	https://regulation.ensembl.org/epigenomes/mus_	2025-12	GRCm39
21	Ensembl_Regulation	atac_seq_forebrain_m_13.5_d.bed.gz	atac_seq_forebrain_m_13.5_d.bb	24/07/2026	Ensembl Regulation	https://regulation.ensembl.org/epigenomes/homo	2025-12	GRCm39
22	Ensembl_Regulation	atac_seq_forebrain_m_13.5_d.chr6.5911252-7710120.bw	atac_seq_forebrain_m_13.5_d.bw	24/07/2026	Ensembl Regulation	https://regulation.ensembl.org/epigenomes/mus_	2025-12	GRCm39
23	Ensembl_Regulation	chip_seq_h3k4me3_forebrain_m_13.5_d.bb	chip_seq_h3k4me3_forebrain_m_13.5_d.bb	24/07/2026	Ensembl Regulation	https://regulation.ensembl.org/epigenomes/mus_	2025-12	GRCm39
24	Ensembl_Regulation	chip_seq_h3k4me3_forebrain_m_13.5_d.chr6.5911252-7710120.bw	chip_seq_h3k4me3_forebrain_m_13.5_d.bw	24/07/2026	Ensembl Regulation	https://regulation.ensembl.org/epigenomes/mus_	2025-12	GRCm39

Assignment

1. Start a clean IGV session.
2. Select/load the correct reference genome (GRCh38/GRCm39/mm10)
3. Load user-defined regions, signal tracks, peak tracks and regulatory elements annotation for the indicated epigenome:
   - Human, HCT116 (male, 48 years old).
   - Mouse, Forebrain E13.5.
4. Rename, reorder, and color tracks by evidence layer, using the following color scale:
   - **ATAC-seq**
   - **DNase-seq**
   - **H3K4me3**
   - **H3K27ac**
   - **H3K4me1**
   - **Custom regions**
5. Set adequate display options for signal and annotation tracks.
6. Inspect the indicated locus and identify:
   - Elements where CRE annotations are concordant.
   - One conflict between annotation or evidence layers.
   - One missing-context example.
   - Locus:
     1. GRCh38, chr8:127,682,026-127,744,101
     2. GRCm39/mm39, chr6:6,740,753-6,891,103; GRCm38/mm10 chr6:6,740,753-6,891,103
7. Save an IGV session, screenshot, or launchable view.